

INTERPOLATION IN BI-INTUITIONISTIC LOGICS

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Interpolation in extensions

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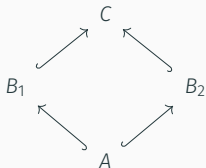
It is a classical logical problem. It can be seen as a **geometric definability theorem**: it concerns the (sub)-definability of **intersections**.

Interpolation and amalgamation

For these logics, interpolation is very closely related to **amalgamation**.

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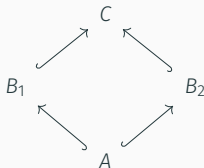
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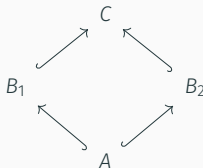
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A class $\mathbf{V}$ has **(super)amalgamation** if each V-formation (super)amalgamates.

In general for our purposes we have the following general equivalence:

Theorem ((Craig) Interpolation = (Super)Amalgamation, Maksimova)

A logic L has:

1. *Deductive interpolation if and only if $\text{Alg}(L)$ has amalgamation;*
2. *Craig interpolation if and only if $\text{Alg}(L)$ has superamalgamation.*

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- 2. Craig interpolation if and only if $\text{Alg}(L)$ has superamalgamation.*

Depends crucially on the congruence extension property/local deduction theorem, and Beth definability properties:

- CRing** has (uniform!) interpolation but fails amalgamation;
- DLat** has a suitable Craig interpolation property, but fails superamalgamation.

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4. **(Categorical logic)**: classify **coreflective subcategories** of V , **regularity of subcategories**, **regularity of monomorphisms**.

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On the other hand:

Theorem (Anti-classification for GL)

*There exist **continuum many extensions** of GL with interpolation.*

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The language: $\mathcal{L}_{\text{biIPC}} = (\wedge, \vee, \rightarrow, \dot{\rightarrow}, 0, 1)$; algebras are **bi-Heyting algebras**, lattices $(H, \leq)$ having, for each $a, b \in H$, elements such that for each c :

$$c \wedge a \leq b \iff c \leq a \rightarrow b$$

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Extensions are closed **super b.i. logics**. A particular example is the biGD logic of Martins and Moraschini:

$$\text{biGD} = \text{biIPC} \oplus (p \rightarrow q) \vee (q \rightarrow p).$$

Its algebras are *bi-Gödel algebras*. It appears naturally in several contexts in the study of trees.

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Today:

1. A classification of super b.i. logics with (Craig/deductive) interpolation **does not seem plausible**.
2. We give a **full classification** for extensions of biGD.

Interpolation in bi-intuitionistic logics

We systematically employ [duality](#).

(Craig) interpolation for bi-intuitionistic logics

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bi-Heyting algebras $\equiv$ Posets with order-topology

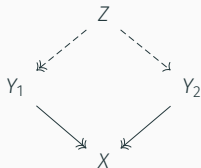
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Co-amalgamation.

A simple example of this:

Theorem

The logic biIPC has the Craig interpolation property.

Proof.

Given $f: Y_1 \rightarrow X$ and $g: Y_2 \rightarrow X$ surjective maps, we take:

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The projections will be bi-p-morphisms, and surjective. Then we use the embedding of $\text{CloUp}(Y_1 \times_X Y_2) \hookrightarrow \text{Up}(Y_1 \times_X Y_2)$, which preserves all existing Heyting and co-Heyting structure. □

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*In $\text{Ext}(\text{bilPC})$ there exist infinitely many tabular logics **with the DIP**.*

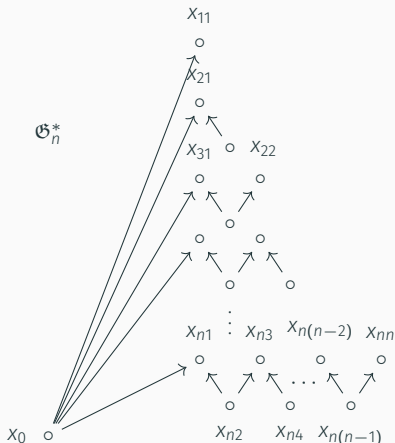
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Question: is there a reasonable combinatorial criterion determining which extensions of bilPC have the CIP?

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Theorem (R.N.A, Qian Chen)

There are exactly six extensions of biGD with DIP:

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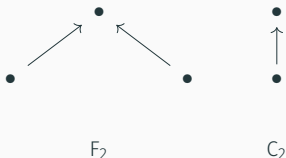
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biLC – logic of *linear frames*; biLC₂ – chains of height at most 2;

biGDBD₂ – logic of co-forks: depth at most 2; biF₂ – logic of the 2-fork.



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For biGD we employ Carai's (2026) representation of **free Gödel algebras**:

Theorem (Carai, 2026)

If X, Y are Esakia spaces dual to Gödel algebras, then the categorical product of them in this category is given by:

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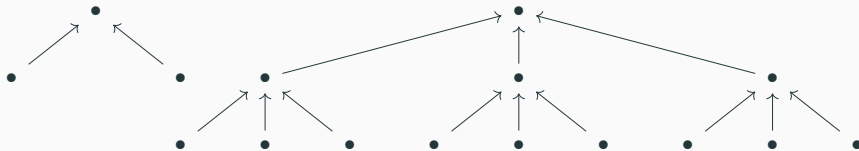
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We describe **categorical pullbacks of Gödel algebras**, and use a similar proof strategy as for biIPC, made more difficult by needing to verify linearity and bi-p-morphism conditions.

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Similar to Maksimova's Jaskowski-like trick: we use [full \$n\$ -ary trees](#).



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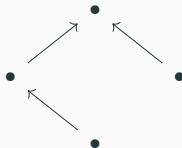
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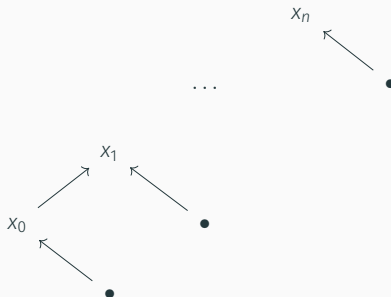
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- If L has depth 1, then it is CPC and we are done.
- If it has depth 2, either it is characterized by C_2 or by F_2 , or it must contain all forks – it must be biGDBD₂.
- If it has depth > 2 and width 1, then it contains a 3-element chain, it must contain all chains – it must be biLC;
- Otherwise, it has depth > 2 and width > 1 : by amalgamation, it contains a *basic 2-comb* Comb₂, and by amalgamation again, all combs Comb _{n} .



n -combs are the structures which generalize the Comb_2 -structure:

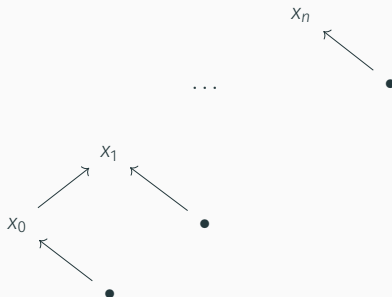


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I.e., LFC is not locally tabular but all of its extensions are.

4. Show that if L contains a 2-comb (or a 3-element chain) and a 3-fork, and has the DIP, then L contains all the frames from $\mathfrak{B}$.

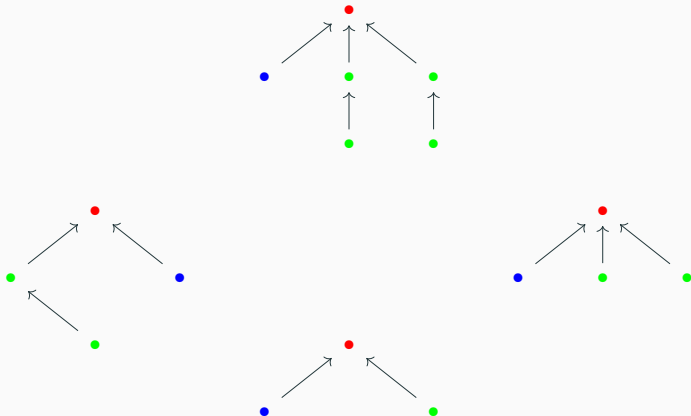
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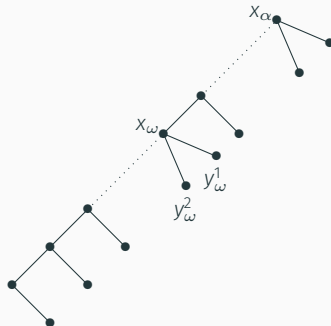
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We need unreasonably large Esakia spaces!

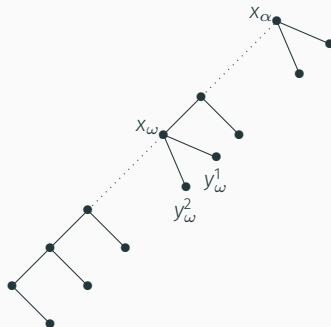
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At limit stages, there are always **two copies on the teeth**. We denote these combs by $\mathfrak{Cb}_\alpha$.

The diagram we consider is then:

 $\mathcal{C}b_{\omega_1}$ $\mathcal{C}b_{\omega}$ 

Where all teeth get identified with the bottom point, and the base goes to the top point.

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Key idea: cofinality argument over ω_1 ensures that we can create the 3-fork.

Final remarks

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2. What about **uniform interpolation**? This fails (in the global sense) for bilPC. We expect it might work for biGD by working over Gödel algebras.

Thank you!
Questions?